

Answer Key — Biology Grade 11

Sub-Strand 2.2: Growth and Development in Plants • Lesson 7: Plant Hormones and Apical Dominance: What Coordinates Growth in Maize and Beans?

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
C	B	C	D	B	B	D	C

USE AFTER: PREDICT PHASE — Predict: Think-Pair-Share on why removing only the tip made the bean branch

Q1. Mr. Wafula's Rosecoco beans with the shoot tip intact had 0 branches at Day 25. Beans with the tip removed at Day 20 had 3 lateral branches by Day 32. What is the best prediction for why removing the tip made the beans branch?

- A. The plant got more sunlight on the side buds after the tip was cut
- B. Cutting the tip made the plant absorb more water from the soil
- C. Something made in the shoot tip was stopping the side buds from growing**
- D. Side buds only grow when the plant is damaged and needs to heal

Correct answer: C. Both beans were the same variety in the same field, so the only difference was the tip. A signal (auxin) from the tip inhibits lateral buds. The sunlight idea is a common misconception, since the lesson controlled other conditions.

USE AFTER: OBSERVE PHASE — Observe: video/storyboard and completing the hormone table (auxin row)

Q2. Which plant hormone is made in the shoot tip, promotes stem elongation and inhibits the growth of lateral buds?

- A. Ethylene
- B. Auxin (IAA)**
- C. Absciscic acid
- D. Cytokinin

Correct answer: B. Auxin (IAA) is made in the shoot tip. It promotes elongation and inhibits lateral buds, which is apical dominance. Cytokinin does the opposite for lateral buds, absciscic acid keeps seeds dormant and ethylene ripens fruit.

USE AFTER: OBSERVE PHASE — Observe: hormone table, ethylene row and oral probe 'Which hormone works as a gas?'

Q3. A learner is completing a hormone table. Which hormone is the odd one out because it works as a gas and promotes fruit ripening?

- A. Gibberellins
- B. Cytokinin
- C. Ethylene**
- D. Auxin (IAA)

Correct answer: C. Ethylene is the gaseous hormone that promotes fruit ripening. The other hormones are not gases and do not mainly ripen fruit.

USE AFTER: EXPLAIN PHASE — Explain: CER on why the Rosecoco bean with tip intact had 0 branches but 3 branches when the tip was removed

Q4. In Mr. Wafula's experiment, a bean plant with its shoot tip intact had 1 main stem and 0 branches at Day 25. A bean plant with the tip removed at Day 20 had 3 lateral branches by Day 32. Which Claim-Evidence-Reasoning statement is the strongest explanation?

- A. The plant with the tip removed was taller, so it grew more branches
- B. Branches grow after Day 30 in all beans, so the tip made no difference
- C. Cytokinin is made in the shoot tip and stops branching, so there were 0 branches
- D. Auxin from the shoot tip inhibits lateral buds; with the tip removed, 3 branches grew instead of 0, so the tip was the cause**

Correct answer: D. A strong CER gives a claim (the tip stops branching), evidence with numbers (0 vs 3 branches) and reasoning naming auxin inhibiting lateral buds. Cytokinin promotes lateral buds, so it does not stop branching.

USE AFTER: EXPLAIN PHASE — Explain: second CER on dormancy of 8 beans, 7 germinated after scarification

Q5. A farmer had 8 healthy-looking bean seeds with hard seed coats that did not germinate. After scarification, 7 of the 8 germinated by Day 7. Which hormone idea best explains why the seeds were dormant before scarification and how they woke up?

- A. Hormones played no part, because only the hard seed coat matters
- B. Absciscic acid maintains dormancy, while gibberellins promote germination**
- C. Ethylene kept the seeds dormant and auxin made them germinate
- D. Cytokinin maintains dormancy, while abscisic acid promotes germination

Correct answer: B. Absciscic acid maintains dormancy and gibberellins promote germination, so hormone balance is involved alongside the hard coat. The other options reverse or misassign the hormone roles.

USE AFTER: DQB PHASE — DQB: sorting questions into Answered today, Partly answered and New

Q6. After the lesson, a learner says: 'Each plant hormone does only one job.' Which fact from this lesson shows that this idea is wrong?

- A. Ethylene promotes fruit ripening
- B. Gibberellins promote both seed germination and stem elongation**
- C. Cytokinins promote cell division
- D. Absciscic acid maintains dormancy

Correct answer: B. Gibberellins have two roles in this lesson: promoting seed germination and promoting stem elongation. Auxin also has two roles, but the other options each give a single role.

USE AFTER: MODEL BUILDING PHASE — Model: revising the Day 1 model with a hormone 'control system' and labelled arrows

Q7. Learners are revising their Day 1 model of maize and beans. Which labelled arrow should they add to show apical dominance?

- A. Absciscic acid from the lateral buds, promoting stem elongation
- B. Gibberellins from the shoot tip, maintaining dormancy
- C. Cytokinin from the shoot tip, inhibiting lateral buds
- D. Auxin from the shoot tip, inhibiting lateral buds**

Correct answer: D. Apical dominance is auxin made in the shoot tip inhibiting lateral buds. Cytokinin promotes lateral buds, abscisic acid maintains dormancy and gibberellins promote germination and elongation.

USE AFTER: END OF LESSON — End of lesson: exit ticket on how the evidence helps answer the driving question

Q8. A farmer in Kitale cuts off the shoot tip of a bean plant and wants more side branches. Which hormone effect explains why this works?

- A. Cutting the tip makes ethylene ripen the side buds into branches
- B. Cutting the tip removes cytokinin, which allows the side buds to grow
- C. Without the tip, less auxin reaches the side buds, so they are no longer inhibited and can grow**
- D. Cutting the tip makes the plant produce more abscisic acid, which grows side buds

Correct answer: C. Auxin from the tip inhibits lateral buds. Removing the tip removes this source, so the lateral buds grow, as in the 0 vs 3 branches data. Cytokinin promotes lateral buds rather than blocking them.